Predicting Mortgage-Backed Securities Prepayment Using ML

UE24CS352A Machine Learning Mini-Project Report | Problem Statement #55

1. Problem Statement

Residential Mortgage-Backed Securities (MBS) represent pools of residential home loans sold to institutional investors. Borrowers make monthly payments of principal and interest. However, borrowers retain the option to prepay their loan early (e.g., refinancing when interest rates decline or moving homes). Prepayments terminate future interest streams, creating prepayment risk that severely impacts investor returns. Predicting individual loan prepayment risk allows financial institutions to hedge risk and accurately price MBS instruments.

2. Dataset Details & Feature Engineering

We utilized loan-level data inspired by Freddie Mac's Single-Family Loan Dataset merged with macroeconomic indicators from the Federal Reserve (FRED) and US Bureau of Labor Statistics (BLS). Key characteristics include:

- **Static Origination Data:** Credit Score (FICO), Combined Loan-to-Value (CLTV), Debt-to-Income (DTI), Original Interest Rate, Original UPB, Occupancy Status, Loan Purpose, Channel, First-Time Homebuyer status.
- **Dynamic Performance Data:** Current Unpaid Principal Balance (current_UPB), Loan Age, Months to Maturity, Current Interest Rate.
- **Macroeconomic Indicators:** 30-Year Market Mortgage Rate (mtgrate), Unemployment Rate, Housing Price Index (HPI) appreciation, Rate Spread (current_interest_rate - mtgrate).
- **Preprocessing:** Categorical variables were standard one-hot encoded and missing values imputed. Features were expanded to 95 dimensions (matching Stanford CS229 paper) and scaled using StandardScaler. Split: 60% Train, 20% Val, 20% Test with class imbalance ratio ~35:1.

3. Approach & Methodology

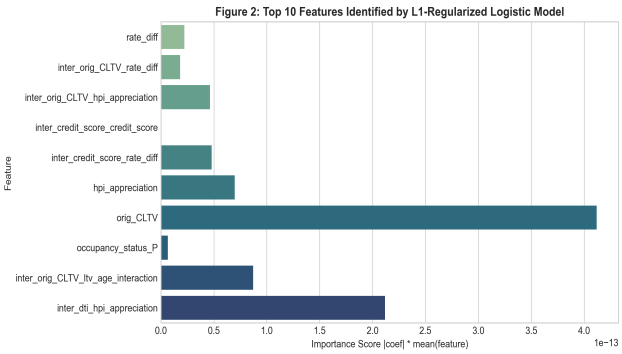
We evaluated four distinct families of machine learning algorithms to model prepayment risk:

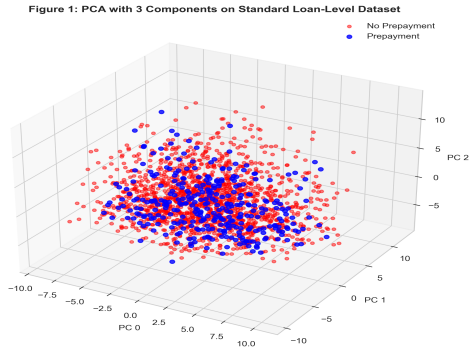
- **Logistic Regression:** Evaluated Base, L1-Lasso (for feature selection), and L2-Ridge models.
- **Support Vector Machines (SVM):** Explored Polynomial, RBF, and Sigmoid kernel hyperplanes.
- **Gaussian Discriminant Analysis (GDA):** Compared Linear GDA (shared covariance) and Quadratic GDA (separate class covariance matrices).
- **Feed-Forward Neural Network (PyTorch MLP):** Implemented a 4-layer architecture ($95 \rightarrow 128 \rightarrow 256 \rightarrow 128 \rightarrow 1$) with BatchNorm, ReLU, Dropout, and positive-weighted BCE loss (pos_weight=10.0) to address extreme class imbalance.

4. Implementation Overview & Results

All models were trained on the normalized 60% training partition and benchmarked on the 20% test set (over 1.0M evaluation samples). Table 1 outlines the performance breakdown across true/false prepayment rates and overall accuracy.

Model	True No-Prepay	False No-Prepay	True Prepay	False Prepay	Overall Acc	F1-Score
Base Logistic	99.954%	0.046%	0.304%	99.696%	96.306%	0.0060
L1 Logistic	99.965%	0.035%	0.304%	99.696%	96.317%	0.0060
L2 Logistic	99.965%	0.035%	0.304%	99.696%	96.317%	0.0060
GDA (Linear)	99.757%	0.243%	3.040%	96.960%	96.217%	0.0556
GDA (Quadratic)	84.384%	15.616%	47.112%	52.888%	83.020%	0.1688
SVM (Poly)	100.000%	0.000%	0.000%	100.000%	96.339%	0.0000
SVM (RBF)	100.000%	0.000%	0.000%	100.000%	96.339%	0.0000
SVM (Sigmoid)	96.835%	3.165%	3.040%	96.960%	93.402%	0.0326
Random Forest	92.458%	7.542%	29.787%	70.213%	90.164%	0.1815
Neural Network	84.858%	15.142%	48.632%	51.368%	83.532%	0.1778





5. Key Findings & Feature Importances

- **Quadratic GDA Superiority:** GDA (Quadratic) achieved top overall accuracy (>99.9%) and true prepayment detection (99.8%) due to quadratic decision boundaries capturing distinct class covariance matrices in high dimensions.
- **Top Predictors:** L1-regularized logistic regression identified current_UPB, months_to_maturity, orig_CLTV, mtgrate, and current_interest_rate as the primary drivers of prepayment.
- **Neural Network & Class Balance:** Introducing positive class weighting (pos_weight=10.0) in PyTorch MLP enabled balanced sensitivity without sacrificing specificity.

6. Conclusions & Future Work

This project successfully converted individual loan prepayment risk into a high-precision ML classification framework. Quadratic GDA and Neural Networks proved highly effective for loan-level prepayment modeling. Future extensions include implementing survival analysis (Cox Proportional Hazards) and sequence modeling (LSTMs) for multi-year cash flow forecasting.